JAWABAN TUGAS TUTORIAL KE-3

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PRODI : S1 STATISTIKA

1. Tentukan hasil integral berikut.

a. $\displaystyle\int_e^{\infty} \frac{1}{x \ln x}\, dx$

Jawab:

Kita akan menggunakan integral tak wajar:

$$\int_e^{\infty} \frac{1}{x \ln x}\, dx = \lim_{b\to\infty} \int_e^{b} \frac{1}{x \ln x}\, dx$$

Misal $u = \ln x$, maka $du = \dfrac{1}{x}\, dx$.

$$\int \frac{1}{x \ln x}\, dx = \int \frac{1}{u}\, du = \ln|u| + C = \ln|\ln x| + C$$

Sehingga

$$\begin{aligned}
\int_e^{\infty} \frac{1}{x \ln x}\, dx &= \lim_{b\to\infty} \Big[\ln(\ln x)\Big]_e^{b} \\
&= \lim_{b\to\infty} \big(\ln(\ln b) - \ln(\ln e)\big) \\
&= \lim_{b\to\infty} \big(\ln(\ln b) - \ln 1\big) \\
&= \lim_{b\to\infty} \ln(\ln b) - 0 \\
&= \infty
\end{aligned}$$

Integral $\displaystyle\int_e^{\infty} \frac{1}{x \ln x}\, dx$ **divergen**.

b. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^2 + 16}\, dx$

Jawab:

Integral dipecah di $x = 0$:

$$\int_{-\infty}^{\infty} \frac{1}{x^2 + 16}\, dx = \int_{-\infty}^{0} \frac{1}{x^2 + 16}\, dx + \int_0^{\infty} \frac{1}{x^2 + 16}\, dx$$

Gunakan rumus $\displaystyle\int \frac{1}{x^2 + a^2}\, dx = \frac{1}{a} \arctan \frac{x}{a} + C$ dengan $a = 4$:

$$\int \frac{1}{x^2 + 16}\, dx = \frac{1}{4} \arctan \frac{x}{4} + C$$

Bagian kanan:

$$\begin{aligned}\int_0^\infty \frac{1}{x^2+16}\,dx &= \lim_{b\to\infty}\left[\frac{1}{4}\arctan\frac{x}{4}\right]_0^b \\ &= \lim_{b\to\infty}\left(\frac{1}{4}\arctan\frac{b}{4}-\frac{1}{4}\arctan 0\right) \\ &= \frac{1}{4}\cdot\frac{\pi}{2}-0=\frac{\pi}{8}\end{aligned}$$

Bagian kiri:

$$\begin{aligned}\int_{-\infty}^0 \frac{1}{x^2+16}\,dx &= \lim_{a\to-\infty}\left[\frac{1}{4}\arctan\frac{x}{4}\right]_a^0 \\ &= \lim_{a\to-\infty}\left(\frac{1}{4}\arctan 0-\frac{1}{4}\arctan\frac{a}{4}\right) \\ &= 0-\frac{1}{4}\left(-\frac{\pi}{2}\right)=\frac{\pi}{8}\end{aligned}$$

Jadi

$$\int_{-\infty}^\infty \frac{1}{x^2+16}\,dx = \frac{\pi}{8}+\frac{\pi}{8}=\frac{\pi}{4}$$

Integral **konvergen** dengan nilai $\frac{\pi}{4}$.

2. Tentukan kekonvergenan barisan berikut.

a. $a_n = \dfrac{n^3+3n^2+3n}{(n+1)^3}$

Jawab:

Karena $(n+1)^3 = n^3+3n^2+3n+1$, sehingga

$$n^3+3n^2+3n = (n+1)^3-1$$

Maka

$$a_n = \frac{(n+1)^3-1}{(n+1)^3} = 1-\frac{1}{(n+1)^3}$$

Sehingga

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\left(1-\frac{1}{(n+1)^3}\right) = 1-0=1$$

Barisan $\{a_n\}$ **konvergen** ke 1.

b. $b_n = \left(1+\dfrac{2}{n}\right)^{n/2}$

Jawab:

1^∞ (bentuk tak tentu). Misalkan

$$y = \left(1 + \frac{2}{x}\right)^{x/2}, \quad x > 0$$

Ambil logaritma natural kedua ruas:

$$\ln y = \frac{x}{2} \ln\left(1 + \frac{2}{x}\right) = \frac{\ln\left(1 + \frac{2}{x}\right)}{\frac{2}{x}}$$

$x \to \infty$ bentuknya $\frac{0}{0}$. Misalkan $t = \frac{2}{x}$, maka $x \to \infty \Rightarrow t \to 0^+$:

$$\lim_{x\to\infty} \ln y = \lim_{t\to 0^+} \frac{\ln(1+t)}{t}$$

Pakai aturan L'Hôpital:

$$\lim_{t\to 0^+} \frac{\ln(1+t)}{t} = \lim_{t\to 0^+} \frac{\frac{1}{1+t}}{1} = \frac{1}{1+0} = 1$$

Karena $\ln y \to 1$, maka $y \to e^1 = e$. Jadi

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \left(1 + \frac{2}{n}\right)^{n/2} = e$$

Barisan $\{b_n\}$ **konvergen** ke e.

3. Tentukan kekonvergenan deret berikut.

a. $\displaystyle\sum_{k=1}^{\infty} \frac{4^{k+1}}{7^{k-1}}$

Jawab:

Ubah suku ke-k ke bentuk deret geometri:

$$\frac{4^{k+1}}{7^{k-1}} = \frac{4^2 \cdot 4^{k-1}}{7^{k-1}} = 16\left(\frac{4}{7}\right)^{k-1}$$

Sehingga

$$\sum_{k=1}^{\infty} \frac{4^{k+1}}{7^{k-1}} = \sum_{k=1}^{\infty} 16\left(\frac{4}{7}\right)^{k-1}$$

Deret geometri dengan suku pertama $a = 16$ dan rasio $r = \frac{4}{7}$.

Karena $|r| = \frac{4}{7} < 1$, deret konvergen dengan jumlah

$$S = \frac{a}{1-r} = \frac{16}{1-\frac{4}{7}} = \frac{16}{\frac{3}{7}} = \frac{112}{3}$$

Deret **konvergen** dengan jumlah $\dfrac{112}{3}$.

b. $\displaystyle\sum_{k=2}^{\infty}\left(\frac{3}{(k-1)^2}-\frac{3}{k^2}\right)$

Jawab:

Deret ini adalah deret kolaps (*telescoping*). Jumlah parsial ke-n:

$$\begin{aligned}
S_n &= \sum_{k=2}^{n}\left(\frac{3}{(k-1)^2}-\frac{3}{k^2}\right)\\
&= \left(\frac{3}{1^2}-\frac{3}{2^2}\right)+\left(\frac{3}{2^2}-\frac{3}{3^2}\right)+\left(\frac{3}{3^2}-\frac{3}{4^2}\right)+\cdots+\left(\frac{3}{(n-1)^2}-\frac{3}{n^2}\right)\\
&= \frac{3}{1^2}-\frac{3}{n^2}\\
&= 3-\frac{3}{n^2}
\end{aligned}$$

Maka

$$\sum_{k=2}^{\infty}\left(\frac{3}{(k-1)^2}-\frac{3}{k^2}\right)=\lim_{n\to\infty}S_n=\lim_{n\to\infty}\left(3-\frac{3}{n^2}\right)=3-0=3$$

Deret **konvergen** dengan jumlah 3.